IESEG School of Management

DSC: YOUR FUNDRASING PARTNER

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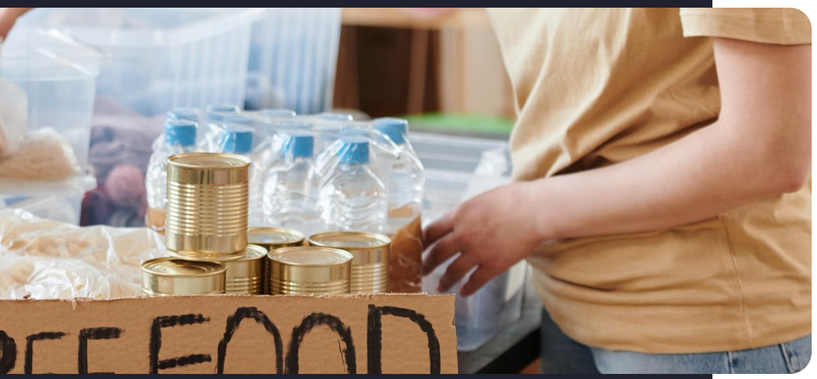
PROJECT OVERVIEW



This project aims to build a predictive model to identify inactive donors who are most likely to donate again during re-activation campaign on 01/02/2020. Using past donor information, campaign history, and donation patterns, the model predicts the probability of a donor contributing at least €30.

- DSC wants to re-activate donors who stopped donating.
- Each letter costs **€0.80.**
- Goal: Maximize campaign profit by ranking potential donors.

DATA

donors

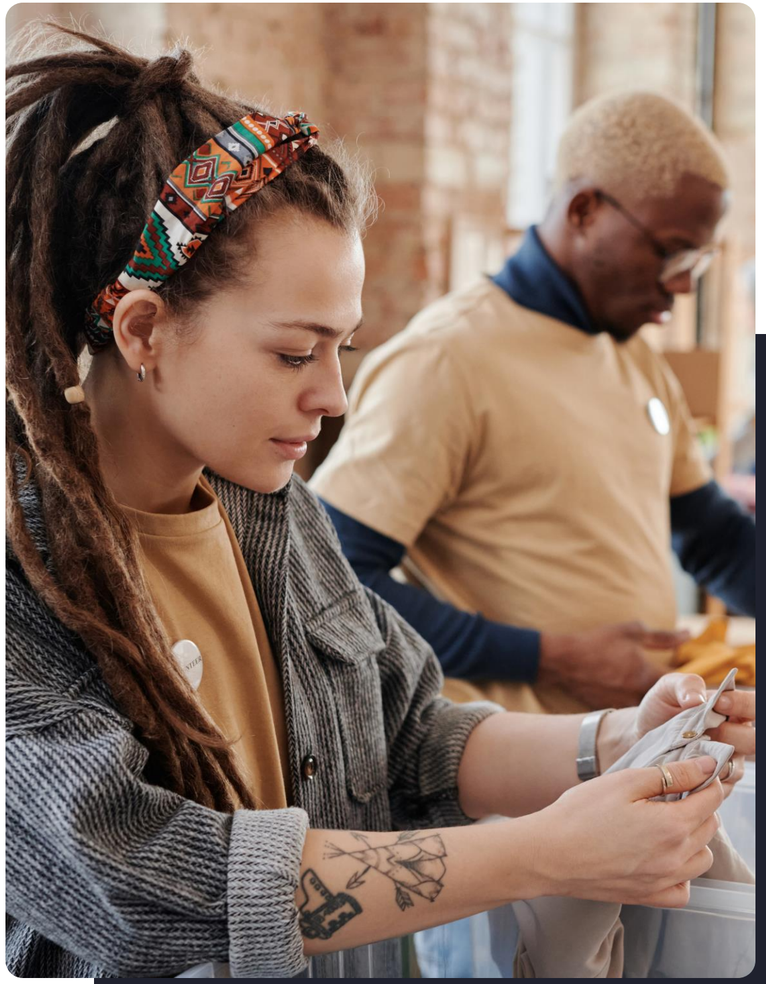
Contains demographic and profile details of each donor, including gender, age, zipcode, province, region, and communication language.

gifts

Records all historical donations made by donors, including donation amount, date, and associated campaign ID

campaigns

Provides information about every campaign launched by DSC, including campaign date, number of letters sent, and cost per letter



DATA

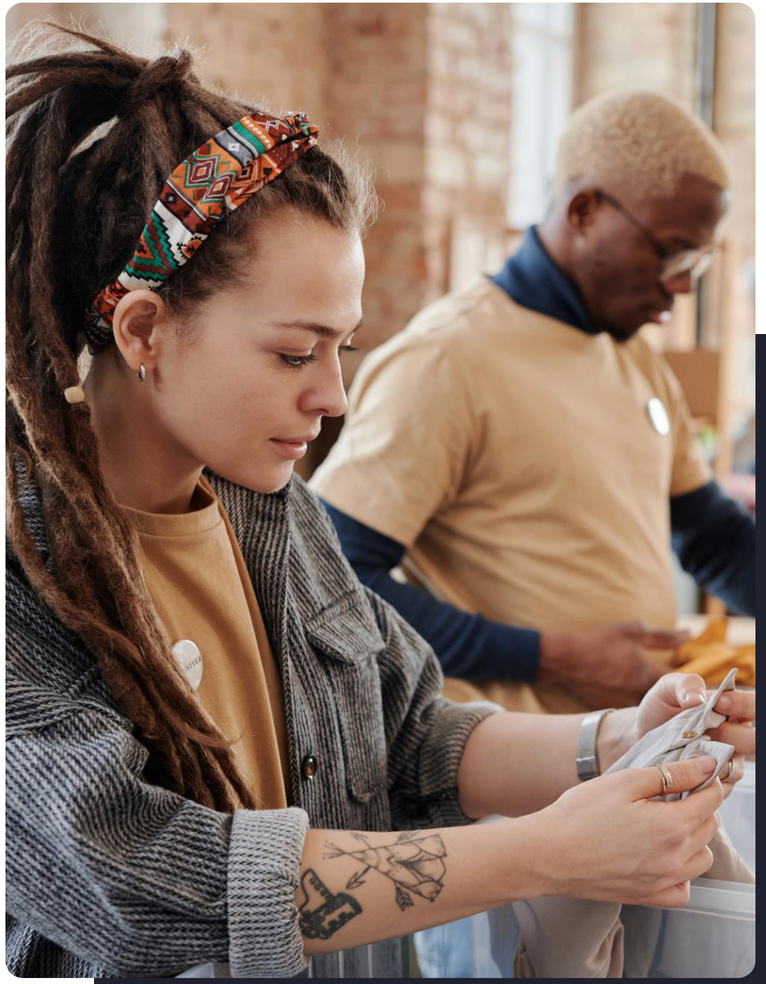
PREVIOUS SELECTION CAMPAIGNS

selection campaign 6169

Donors that were selected for a previous re-activation campaign on 04/09/2018

selection campaign 7244

Donors that were selected for a previous re-activation campaign on 18/06/2019



TIMELINE

Current Campaign (7362) Drop Date: 01/02/2020

Gap definition: 7 days before campaign

TRAINING DATA

Strict time window of 5 years to train our model on the 6169 campaign:
From 27/08/2013
To 28/08/2018

TESTING DATA

Strict time window of 5 years to test our model on the 7244 campaign:
From 09/06/2014
To 11/06/2019

SCORING DATA

Strict time window of 5 years to score our model on the 7362 campaign:
From 23/01/2015
To 23/01/2020

Feature Engineering

Demographics

- gender
- province
- region
- language
- age
- age group

Engagement

- Donation frequency
- Most recent donation
- years as donor
- Frequency of indep. donations
- times donated during campaigns

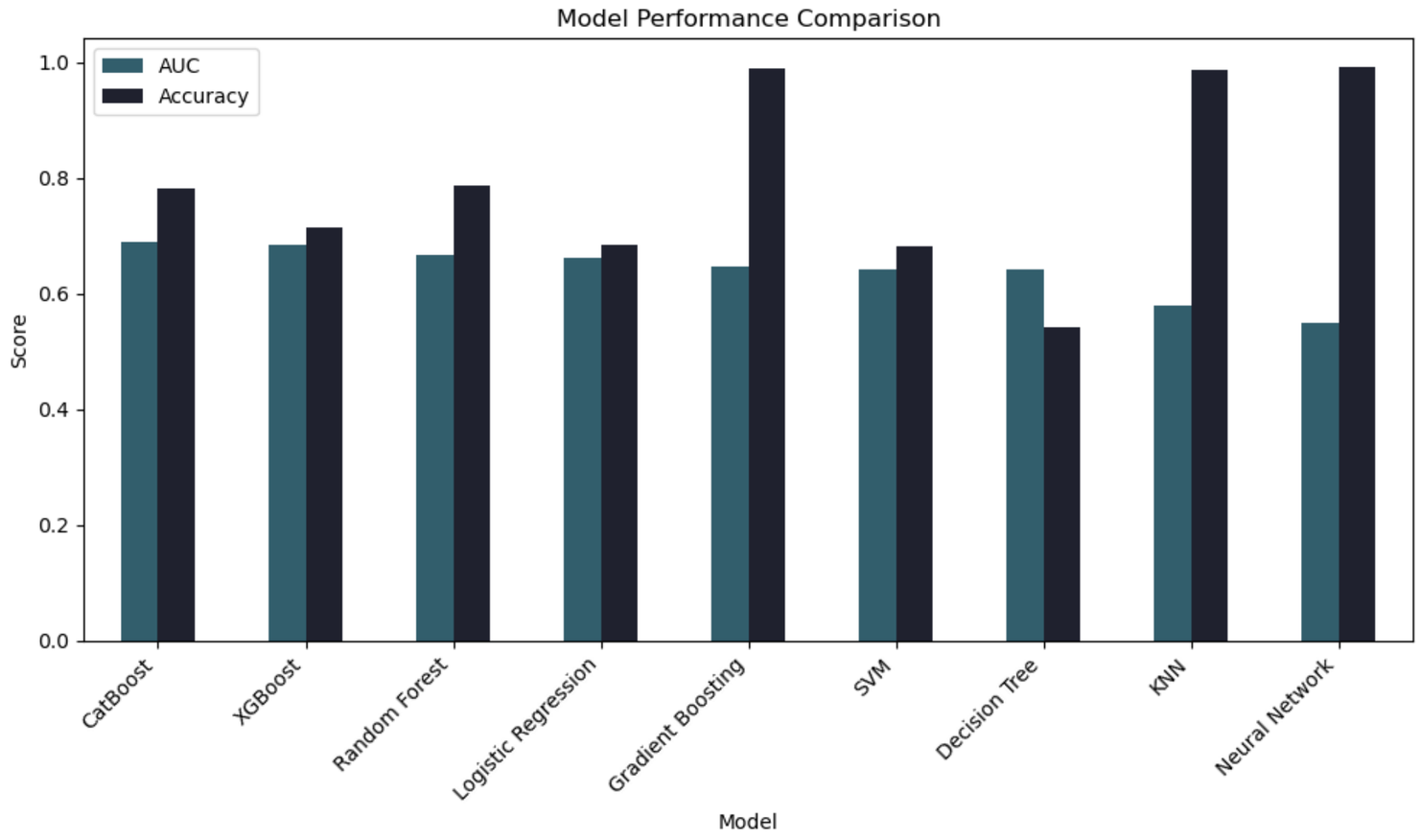
Financial

- Total amount given
- Avg. Amount given
- Max. Amount given
- Min. Amount given
- Last gift amount
- Volatility of donations

Flags

- is independent donor
- is top 10% of donors
- is top 20% of donors

FINDING THE BEST MODEL



Tested 9 different algorithms

prioritized AUC (Area Under Curve) over Accuracy due to imbalance in data (very few donors in proportion to total population)

Best model: **CatBoost**
Accuracy: **78.21%**
AUC: **0.6905**

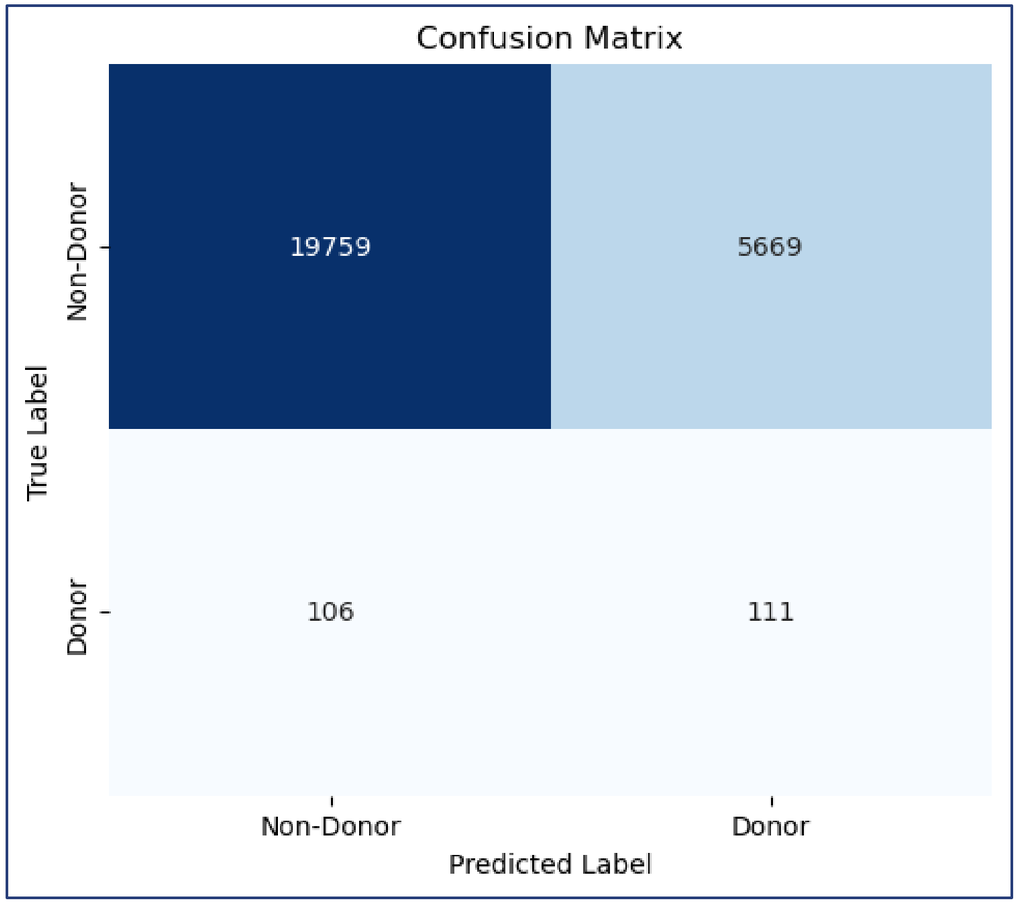
OPTIMIZING CATBOOST

Iterations	Number of trees to build →	**100**
Depth	The depth of the trees →	**4**
Learning Rate	Step size shrinkage to prevent overfitting →	**0.1**
L2 Leaf Reg	Regularization term to smooth the model →	**3**
Scale Pos Weight	Based on the imbalance of our training data →	**~35.7**

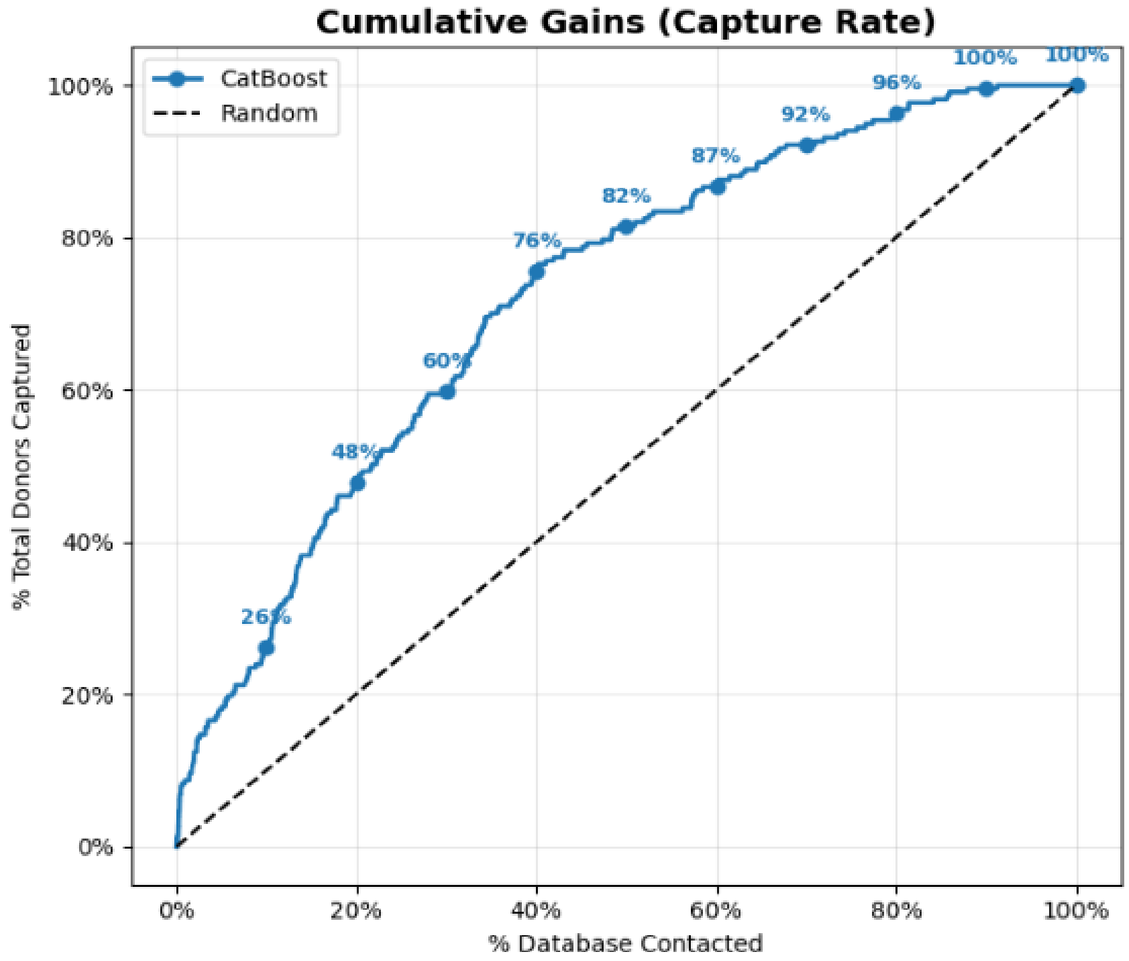
FINAL PERFORMANCE METRICS

AUC: 0.73

Accuracy: 77.5%



MODEL EVALUATION: Cumulative Gains

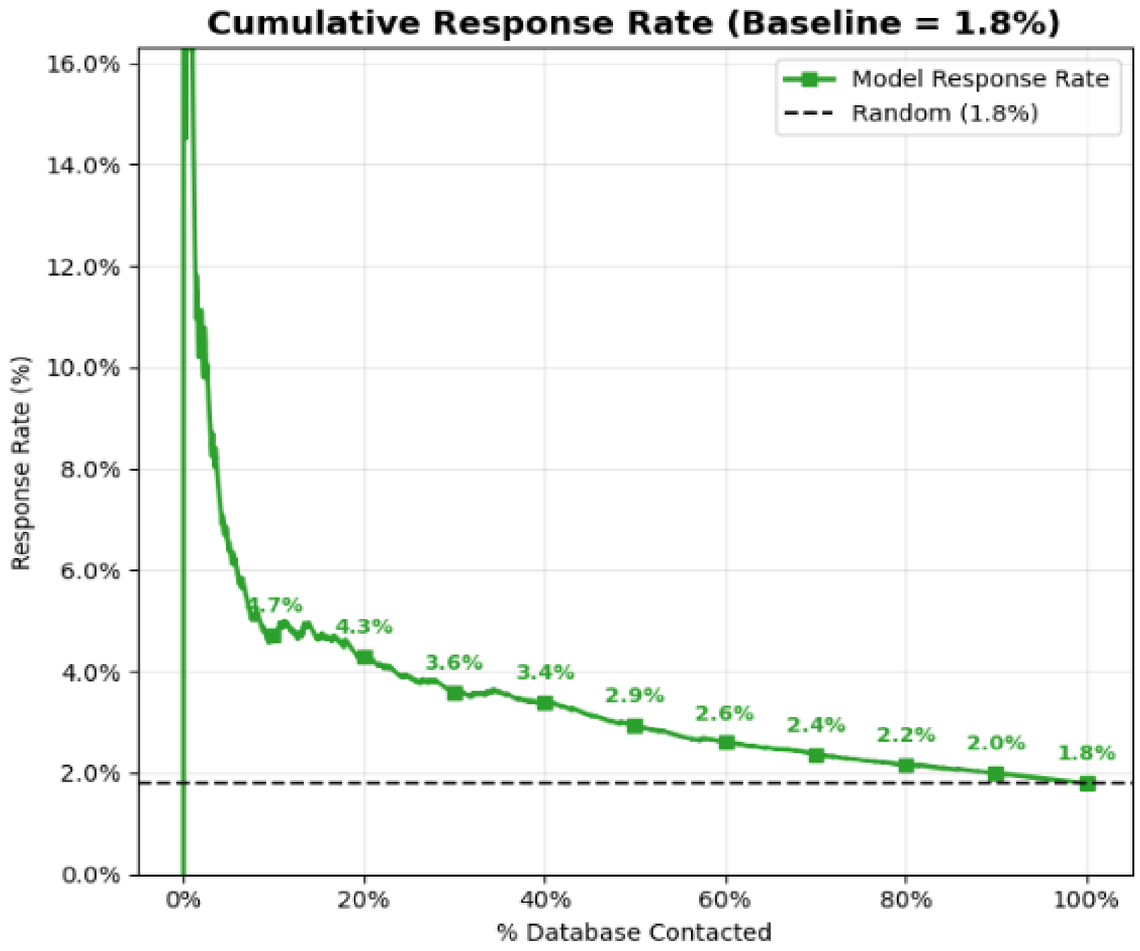


Observation:

The CatBoost model captures donors much faster than random.

- Contacting the top 20% of the database will capture ~48% of responding donors.
- Contacting the top 40% reaches ~76% responding donors.
- Contacting 100% reaches 100% (as expected).

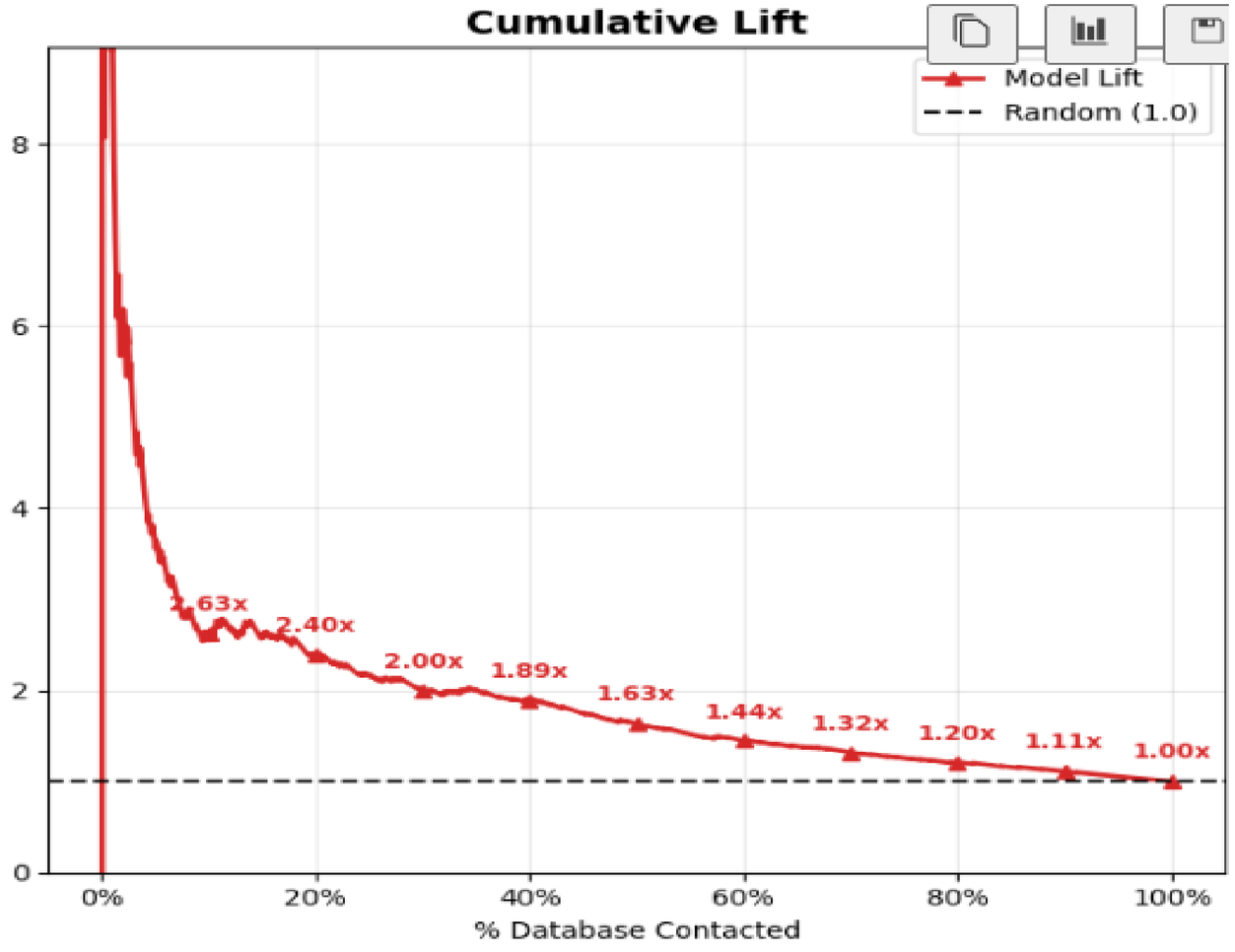
MODEL EVALUATION: Cumulative Response Rate



Observation:

- The response rate is highest at the beginning (~16% for the top-ranked donors) and gradually decreases as lower-ranked donors are contacted.
- Even at ~40% of the database contacted, the response rate (~3.4%) is still higher than random.

MODEL EVALUATION: Cumulative Lift



Observation:

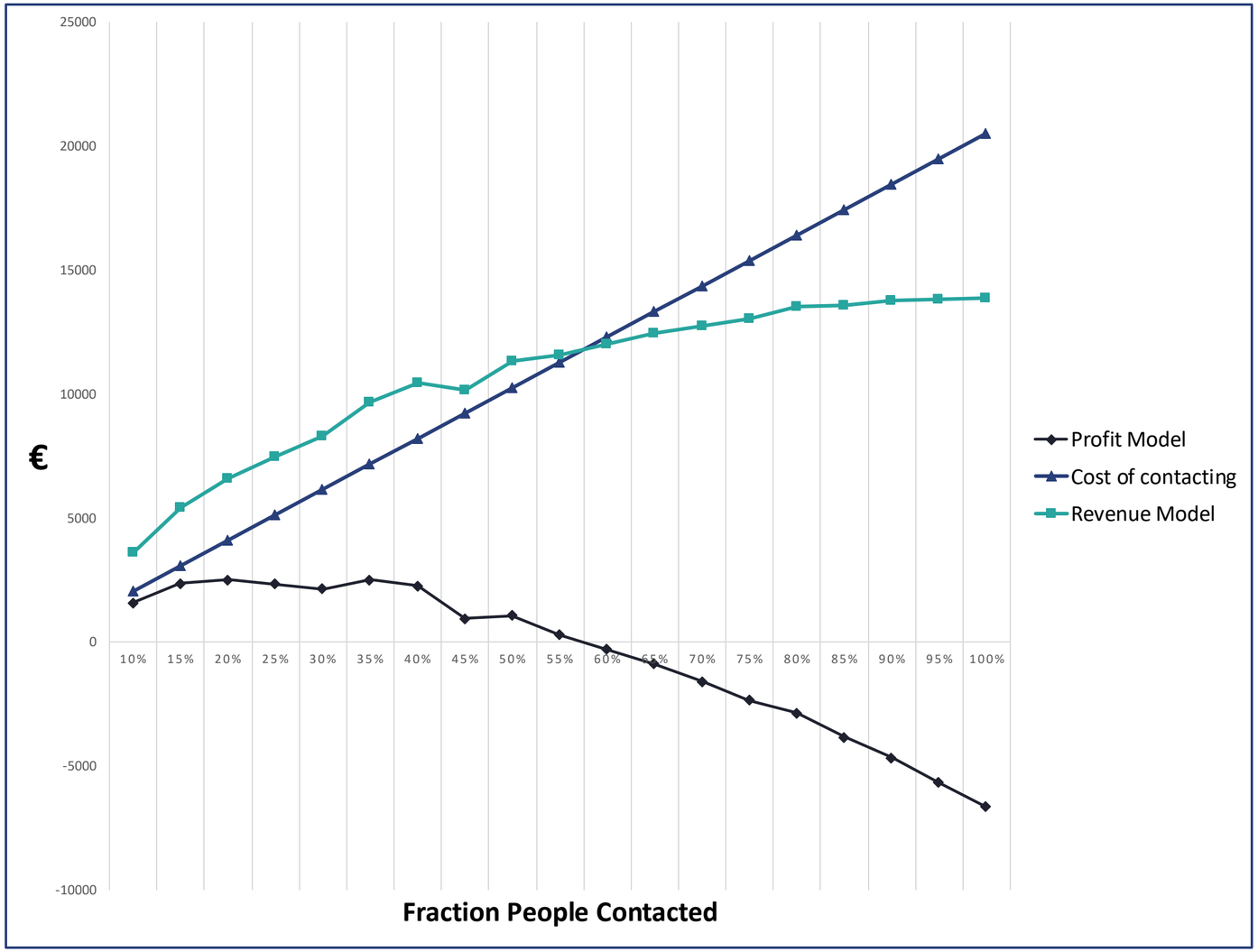
- The model lift is highest for the top-ranked donors (up to ~9x), meaning these donors are 9 times more likely to respond than a random donor.
- The lift gradually decreases as more of the database is contacted, eventually reaching 1x (same as random).

OUTCOMES USING PREDICTIVE MODEL

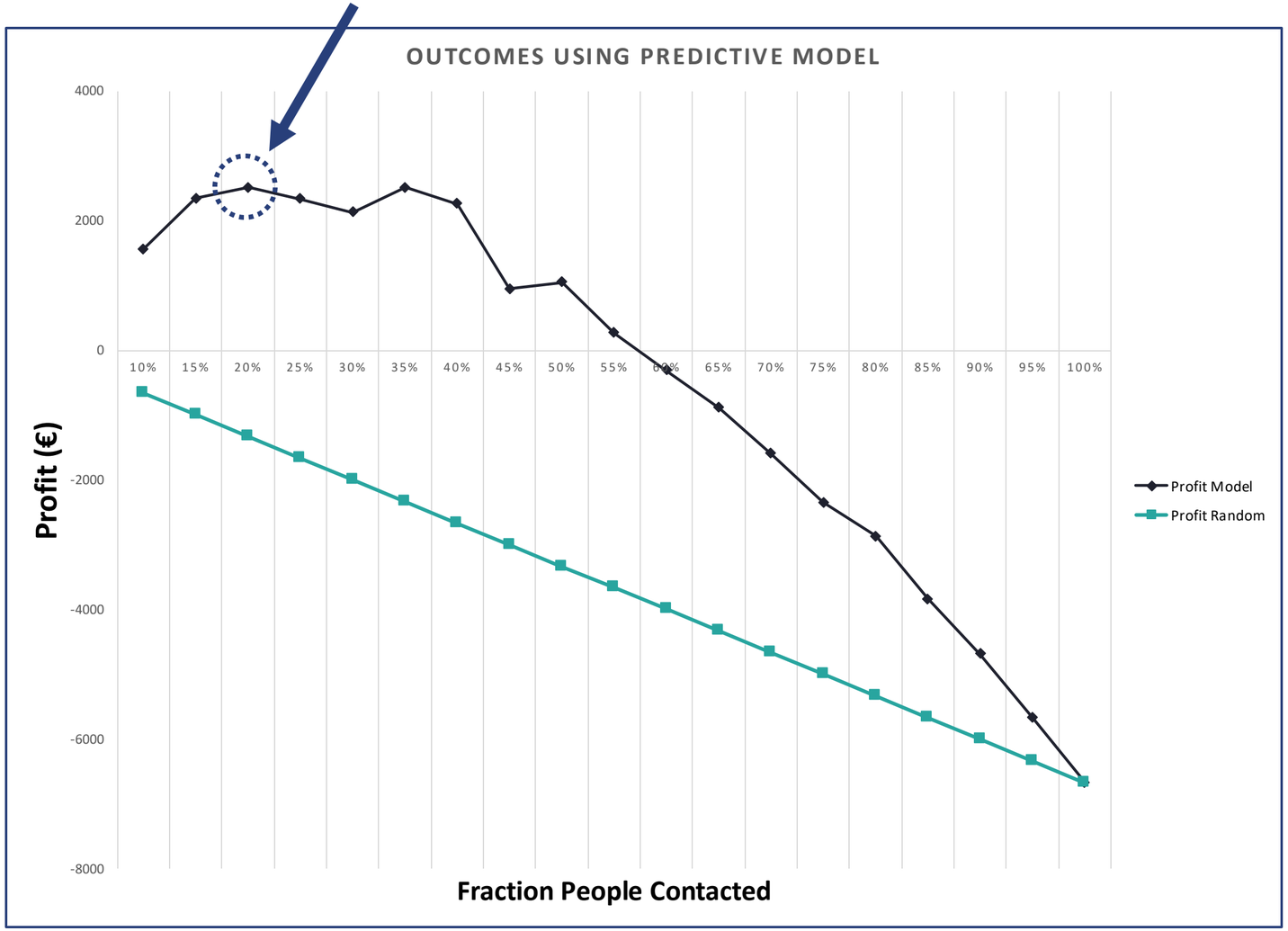
ASSUMPTIONS:

- AVERAGE DONATION OF 30€ (WORST CASE SCENARIO)
- COST OF CONTACTING OF 0.8 € PER LETTER SENT

Model generates positive profit up to about 55%–60%



PROFIT FROM MODEL VS RANDOM SELECTION

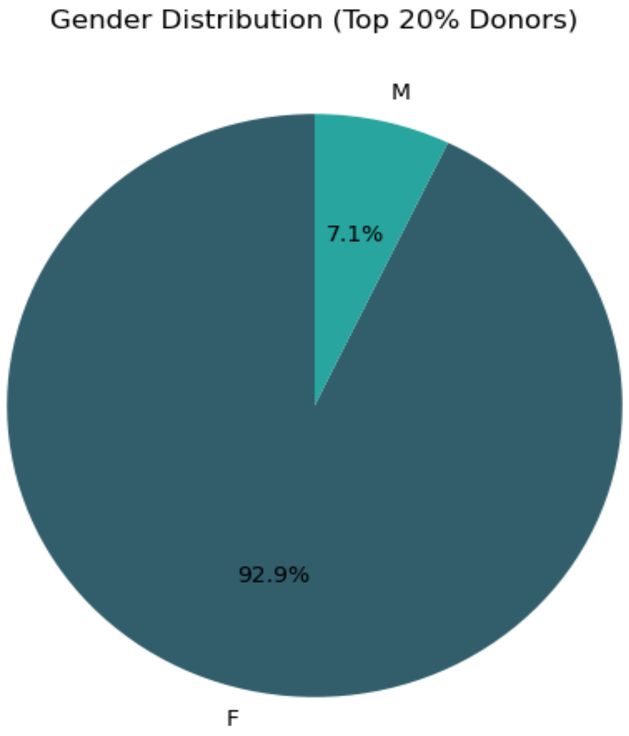


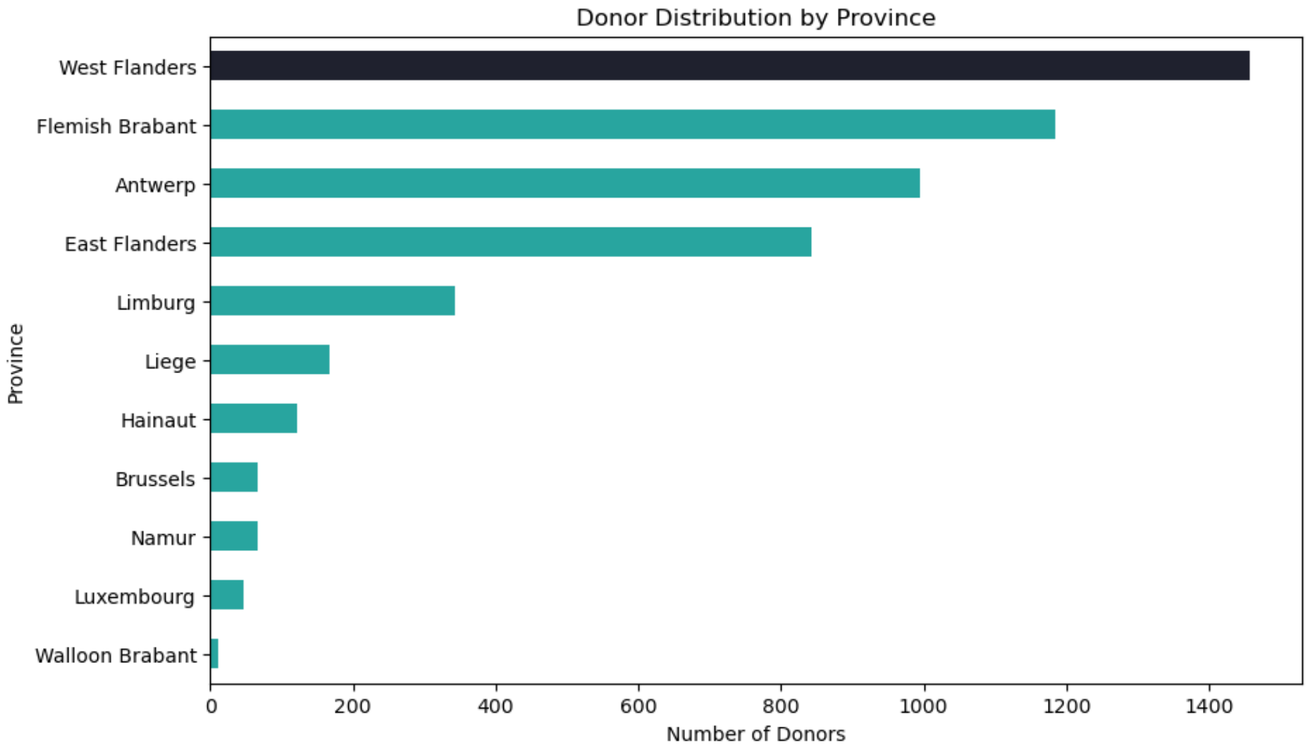
PROFIT MAXIMISES WHEN CONTACTING THE TOP **20%** OF THE DONOR SET

MODEL CONCLUSIONS

Fraction of people to be contacted	Total cost of the campaign	Expected Campaign Revenue	Expected Campaign Profit	ROI
20%	4,103€	6,616€	2,513€	61%

Donor Profiles and Characteristics





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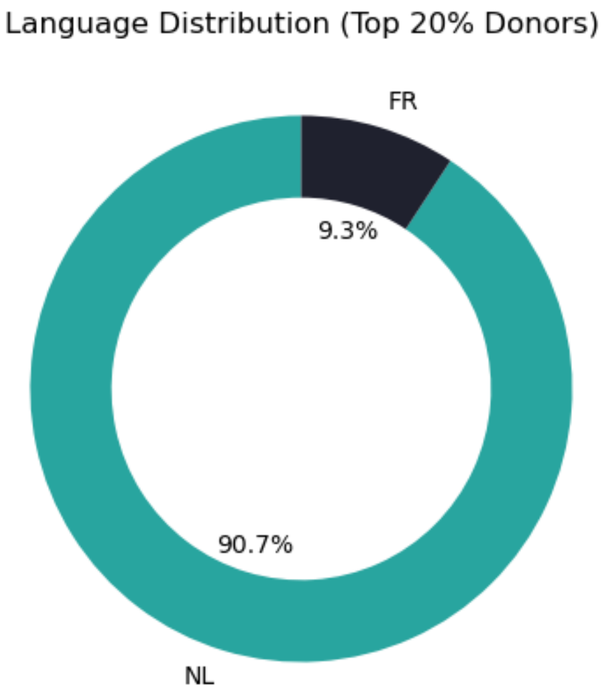
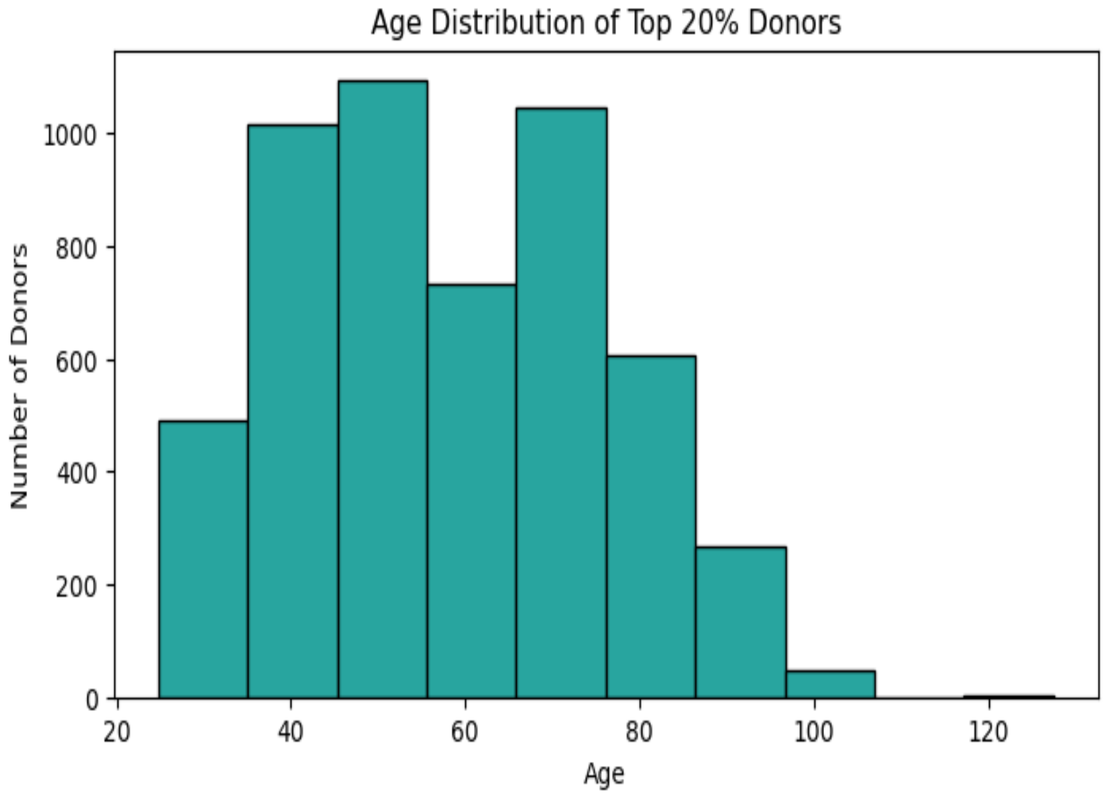




Among the top 20% of donors, most are female, primarily from West Flanders, speak Dutch (NL), and are aged between 50 and 60

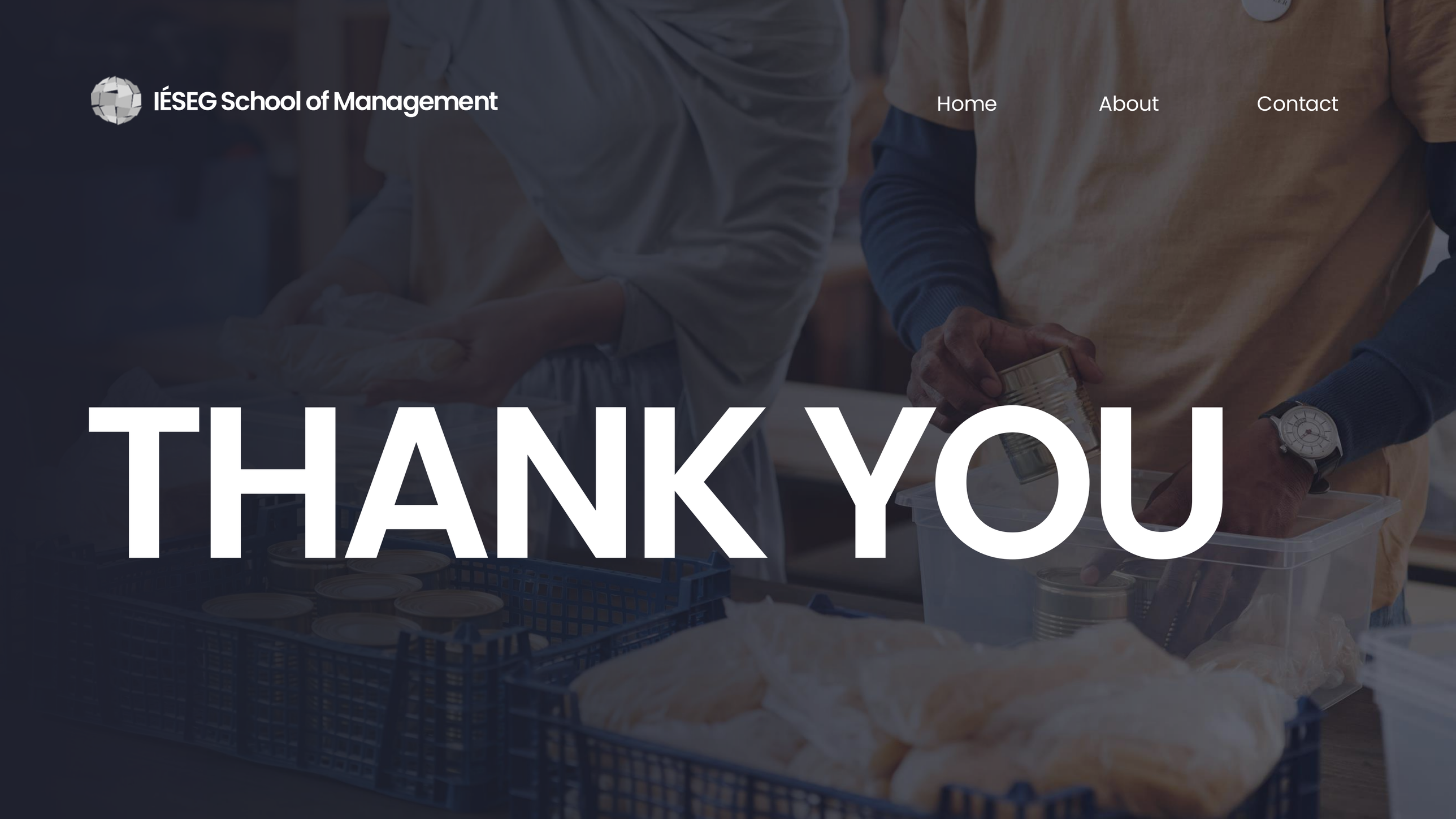
RECOMMENDATIONS

1. **Contact only the top 20% of customers** ranked by the predictive model, as this is where profit is maximized.

2. **Avoid contacting more than 40%** of the customer base, because costs begin to outweigh revenue and profits turn negative.

3. **Do not use a random-contact strategy**, since it produces losses at every contact level and wastes budget.

4. **Use the predictive model to prioritize high-probability responders**, as it significantly improves response rate and revenue per contact.



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THANK YOU